Introduction to Detection Theory

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Key Concepts

1) Detection theory is concerned with using a test to decide which of several hypotheses produced the observed data.

2) Categories of tests and hypotheses:

    a) Binary tests are concerned with comparing two hypotheses about the data, such as the presence of cancer vs the absence of cancer.

    b) M-ary tests are concerned with comparing more than two hypotheses about the data.

c) Simple hypotheses are those for which the probability density function of the data is completely known.

d) Composite hypotheses are those for which the probability density function has unknown parameters, such as noise power or signal amplitude.

3) Binary tests usually involve comparing a function of the data to a threshold.

4) Binary hypothesis test performance characterization

a) Probability of "detection" (sensitivity) is the probability of deciding H_1 when H_1 is the true underlying hypothesis.

b) Probability of "false alarm" (type I error) is the probability of deciding H_1 when H_0 is the true underlying hypothesis.

c) Probability of "miss" (type II error) is the probability of deciding H_0 when H_1 is the true underlying hypothesis.

d) Specificity is the probability of deciding H_0 when H_0 is the true underlying hypothesis.

e) Notions of detection, false alarm, and miss are based on H_0 being noise and H_1 being a the signal due to an object + noise.

f) The receiver operating characteristic (ROC) is a plot of

the probability of detection as a function of the probability of false alarm as the test threshold is varied.

i) Equivalently, the ROC is a plot of the sensitivity as a function of 1-specificity as the test threshold is varied.

ii) The test that lies furthest into the upper left corner of the ROC is the better test when comparing tests.

Video Screenshots

Given data, decide if an event occurs

Test classification

1) Binary: two possibilities

Example - signal present vs signal absent

H_0: $x[n] = w[n]$ (noise)

H_1: $x[n] = s[n] + w[n]$ (signal plus noise)

2) M-ary, multiple test: more than two possibilities

Example - classification

H_i: $x[n] = s_i[n] + w[n]$, $i = 1, 2, \ldots, M$

which signal is it?

1:58

Types of hypotheses

1) Simple H_i : data x has a fully known pdf

Example: $\underline{x} \sim N(\underline{\mu}_i, \underline{R})$ with $\underline{\mu}_i$, $\underline{R}$ known

2) Composite H_i : pdf of $\underline{x}$ is not fully known

Example: $\underline{x} \sim N(A\underline{s}, \sigma^2 \underline{\underline{I}})$ A, σ^2 unknown

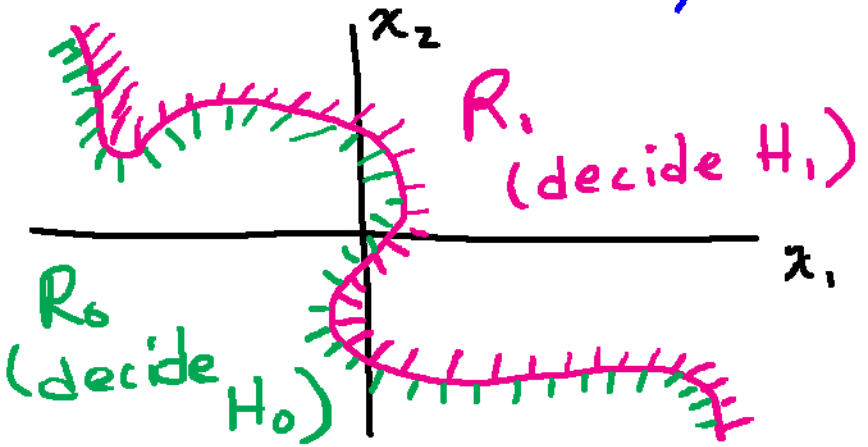
signal $A\underline{s}$ plus noise

Goal is to observe $\underline{x}$, decide which is the best hypothesis

Let $R_0 = \{\underline{x} : \text{decide } H_0\}$

$R_1 = \{\underline{x} : \text{decide } H_1\}$

$\underline{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$



Performance characterization (Binary)

Four possibilites

1) "Detection" H_1 true, decide H_1 — Prob{H_1 true, decide H_1} = P_D
   - sensitivity -

2) "False Alarm" H_0 true, decide H_1 — Prob{H_0 true, decide H_1} = P_{FA}
   - type I error -

3) "Miss" H_1 true, decide H_0 — Prob{H_1 true, decide H_0} = $P_M = 1 - P_D$
   - type II error -

4) H_0 true, decide H_0 — Prob{H_0 true, decide H_0} = $1 - P_{FA}$
   - specificity -

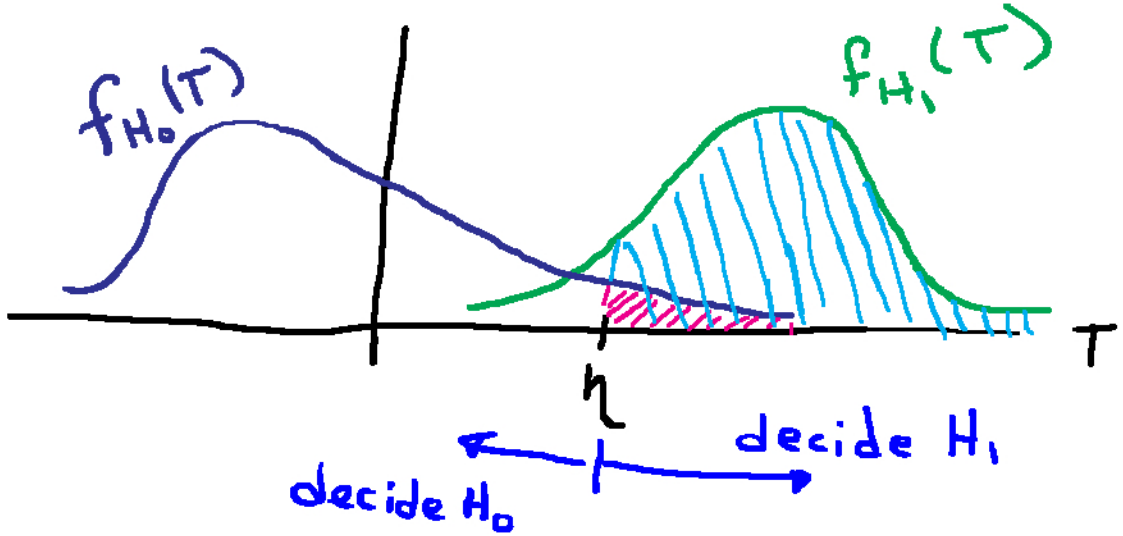
$$P_{FA} = \int_{x \in R_1} f(\underline{x} \mid H_0 \text{ true}) \, d\underline{x} \qquad P_D = \int_{x \in R_1} f(\underline{x} \mid H_1 \text{ true}) \, d\underline{x}$$

Typically regions R_0, R_1 are defined using a function and threshold

$$T(\underline{x}) \underset{R_0}{\overset{R_1}{\gtrless}} \eta \Rightarrow T(\underline{x}) \underset{H_0}{\overset{H_1}{\gtrless}} \eta$$

$f_{H_0}(T)$: pdf of $T(\underline{x})$ when H_0 is true

$f_{H_1}(T)$: pdf of $T(\underline{x})$ when H_1 is true



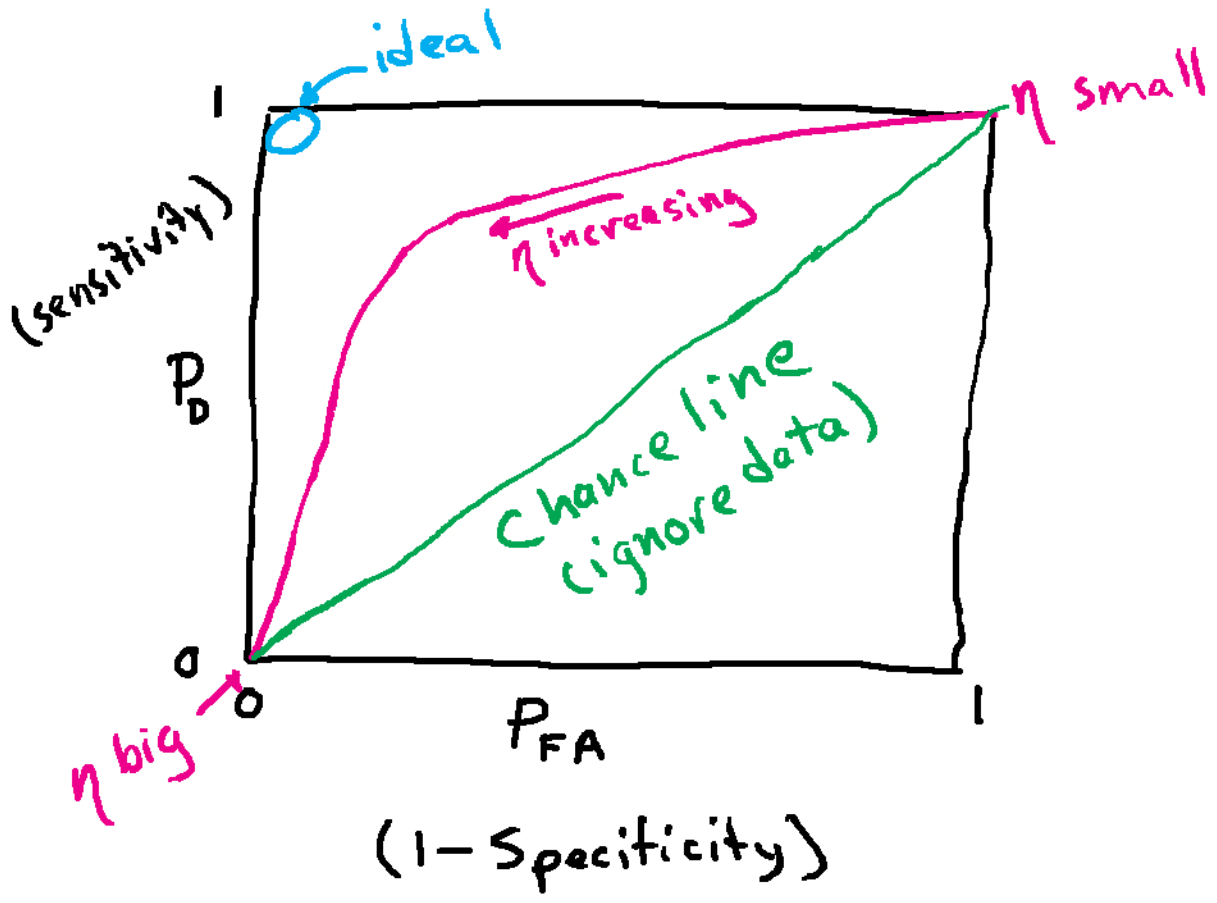
$$P_{FA} = \int_{\eta}^{\infty} f_{H_0}(T)\,dT$$

$$P_D = \int_{\eta}^{\infty} f_{H_1}(T)\,dT$$

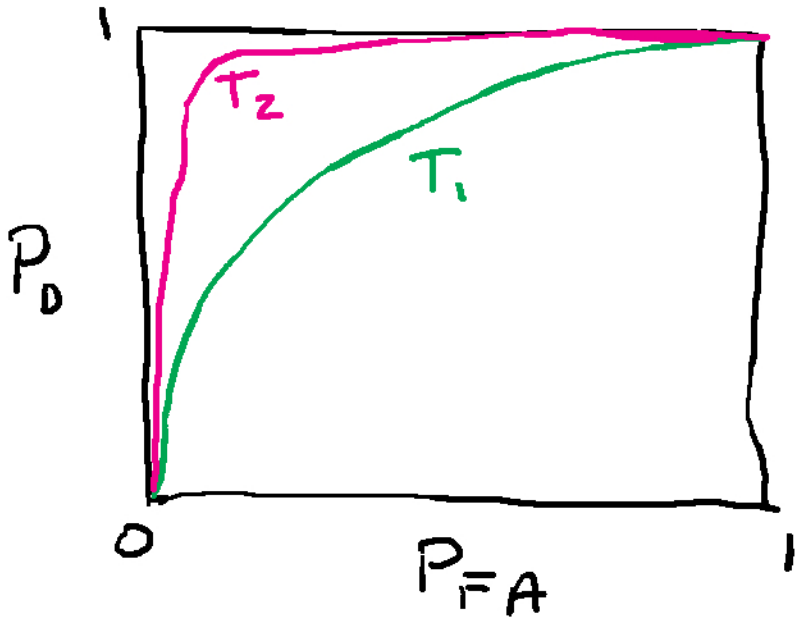
As $P_{FA} \downarrow$ ($\eta \uparrow$), P_D also $\downarrow$

Receiver Operating Characteristic (ROC)

- describes tradeoff between P_{FA} and P_D for a test $T(\underline{x})$



Comparing tests



T_2 is the better test

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